A 7/27 BOUND FOR PENNY GRAPHS

Abstract. A Euclidean minimum-distance graph is obtained from a finite point set in the plane whose mutual distances are at least one by joining pairs at distance exactly one. We prove that every such graph on n vertices has an independent set of size at least $\lceil 7n/27 \rceil$. This improves the lower bound $8n/31$ of Swanepoel. The proof combines a minimal-counterexample expansion argument, a self-contained boundary reduction adapted from the work of Csizmadia and Swanepoel, and a finite elimination of the resulting Euclidean boundary configurations.

1. Introduction

Let $P \subset \mathbb{R}^2$ be finite and suppose that $|x - y| \geq 1$ for distinct $x, y \in P$. Its *minimum-distance graph* $G(P)$ has vertex set P, with an edge between two points precisely when their distance is one. Equivalently, these are the contact graphs of packings of congruent closed discs. Write

$$g(n) = \min\{\alpha(G(P)) : |P| = n\},$$

where α denotes the independence number.

Pollack observed that planarity gives $g(n) \geq n/4$ [3]. Csizmadia improved this to $9n/35$ [1], and Swanepoel obtained $8n/31$ [4]. The best known construction gives the upper bound $5n/16 + O(1)$ [2]. We prove the following.

Theorem 1.1. *Every Euclidean minimum-distance graph G satisfies*

$$\alpha(G) \geq \left\lceil \frac{7|V(G)|}{27} \right\rceil.$$

Consequently, $g(n) \geq \lceil 7n/27 \rceil$ for every n.

The proof has three parts. First, a smallest counterexample must expand every small independent set: an independent set of at most seven vertices has at least three times as many neighbours. A boundary-curvature argument then forces a long outer-boundary chain consisting almost entirely of equilateral triangles and degree-four vertices. Second, Euclidean geometry shows that the next layer of this chain agrees with the triangular lattice except at one prescribed break. Finally, every possible continuation of the chain contains an independent set whose neighbourhood is too small, contradicting minimality.

The boundary reduction is adapted from the method introduced by Csizmadia and sharpened by Swanepoel, but all ingredients needed here are proved below. The specialization to the Euclidean plane and to the constant 7/27 permits a substantially shorter geometric description than the general normed-plane argument.

Throughout, graphs are finite and simple. For $S \subseteq V(G)$, let

$$N_G(S) = \{v \in V(G) \setminus S : v \text{ has a neighbour in } S\}.$$

We omit the subscript when the graph is clear.

2. Minimal counterexamples and boundary curvature

Assume for contradiction that Theorem 1.1 fails, and choose a counterexample G with the fewest vertices. Put $n = |V(G)|$.

Lemma 2.1 (Deletion criterion)**.** *If $\varnothing \neq S \subseteq V(G)$ is independent and*

$$7|N(S)| \leq 20|S|,$$

then G is not a counterexample.

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Proof. Delete $S \cup N(S)$, and let $G' = G - (S \cup N(S))$. The graph G' is a minimum-distance graph on fewer vertices, so minimality gives $\alpha(G') \geq 7|V(G')|/27$. Since no vertex of G' is adjacent to S,

$$\alpha(G) \geq |S| + \alpha(G') \geq |S| + \frac{7}{27}\big(n - |S| - |N(S)|\big) = \frac{7n + 20|S| - 7|N(S)|}{27} \geq \frac{7n}{27},$$

a contradiction. □

Corollary 2.2 (Small-set expansion)**.** *Every independent set S with $1 \leq |S| \leq 7$ satisfies $|N(S)| \geq 3|S|$. In particular, G has minimum degree at least three.*

Proof. If $|S| = k \leq 7$ and $|N(S)| \leq 3k-1$, then $7|N(S)| \leq 21k-7 \leq 20k$, contrary to Lemma 2.1. The final assertion is the case $k = 1$. □

Lemma 2.3. *The straight-line drawing of a Euclidean minimum-distance graph is planar, and every vertex has degree at most six.*

Proof. Suppose two unit segments cross properly at z. One endpoint x of the first segment and one endpoint y of the second satisfy $|x - z| \leq 1/2$ and $|y - z| \leq 1/2$. Thus $|x - y| \leq 1$, and equality would force the two segments to be collinear, contrary to proper crossing. Hence $|x - y| < 1$, impossible.

The neighbours of a vertex lie on its unit circle. If two such neighbours subtend an angle $\theta \in [0, \pi]$, then their distance is $2\sin(\theta/2)$, so $\theta \geq \pi/3$. At most six neighbour directions fit around the circle. □

Lemma 2.4. *The graph G is two-connected. Consequently, the boundary of its outer face is a simple cycle.*

Proof. If G were disconnected, the union of independent sets supplied by minimality in its components would have size at least $7n/27$.

Suppose that x is a cut vertex. Partition the components of $G - x$ into two nonempty unions with vertex sets A and B, and put $H_A = G[A \cup \{x\}]$, $H_B = G[B \cup \{x\}]$. Since adding one vertex changes the independence number by at most one, either $\alpha(H_A) = \alpha(G[A])$ or $\alpha(H_A) = \alpha(G[A]) + 1$.

In the first case, a maximum independent set of H_A can be chosen to avoid x. Combining it with an independent set in $G[B]$, and applying minimality to both graphs, gives at least $7(|A| + 1)/27 + 7|B|/27 = 7n/27$ vertices.

In the second case, let I_A be a maximum independent set of H_A containing x. Then $I_A \setminus \{x\}$ is a maximum independent set of $G[A]$ and contains no neighbour of x. It can therefore be combined with any independent set of H_B, even one containing x. Minimality gives at least $7|A|/27 + 7(|B| + 1)/27 = 7n/27$ vertices. Both cases are contradictory. □

We use two elementary angle facts repeatedly. Angles below are the ordinary Euclidean angles in $[0, \pi]$.

Lemma 2.5 (Three-edge angle inequality)**.** *Let a, b, c, d occur in this order on a simple quadrilateral, and suppose*

$$|a - b| = |b - c| = |c - d| = 1, \qquad |a - c|, |a - d|, |b - d| \geq 1.$$

Then

$$\angle abc + \angle bcd \geq \pi.$$

Proof. Put $b = (0, 0)$, $c = (1, 0)$, and write $\beta = \angle abc$, $\gamma = \angle bcd$. The inequalities $|a - c| \geq 1$ and $|b - d| \geq 1$ imply $\beta, \gamma \geq \pi/3$. After reflecting in the line bc, we may take $a = (\cos\beta, \sin\beta)$.

Suppose first that a and d lie on the same side of the line bc. Then $d = (1 - \cos\gamma, \sin\gamma)$. If $\beta + \gamma < \pi$, then

$$|a - d|^2 = 3 + 2\cos(\beta + \gamma) - 2\cos\beta - 2\cos\gamma < 1,$$

because $\cos\beta + \cos\gamma > 1 + \cos(\beta + \gamma)$. This contradicts $|a - d| \geq 1$.

Suppose instead that a and d lie on opposite sides of bc, so $d = (1-\cos\gamma, -\sin\gamma)$. If $\beta+\gamma < \pi$, the segment ad meets the line bc at abscissa

$$\xi = \frac{\sin\gamma\cos\beta + \sin\beta(1-\cos\gamma)}{\sin\beta + \sin\gamma}.$$

Its numerator and the difference between its denominator and numerator are

$$2\sin\frac{\gamma}{2}\cos\left(\beta - \frac{\gamma}{2}\right) \quad \text{and} \quad 2\sin\frac{\beta}{2}\cos\left(\gamma - \frac{\beta}{2}\right),$$

respectively. They are positive because $\beta, \gamma \geq \pi/3$ and $\beta + \gamma < \pi$. Thus $0 < \xi < 1$, so ad crosses the open segment bc, contrary to simplicity. Hence in every case $\beta + \gamma \geq \pi$. □

Lemma 2.6 (Cycle curvature)**.** *Let Q be a simple cycle in a minimum-distance graph, and let H be the subgraph induced by the vertices on or inside Q. If D_j is the number of vertices of degree j in H on Q, then*

$$\sum_{j\geq 2}(4-j)D_j \geq 6.$$

In particular, $2D_2 + D_3 \geq 6$.

Proof. At a boundary vertex of degree j in H, the interior angle of Q contains $j-1$ successive gaps between neighbour directions, each at least $\pi/3$. If Q has t vertices, its angle sum is $(t-2)\pi$, whence

$$(t-2)\pi \geq \frac{\pi}{3}\sum_{j\geq 2}(j-1)D_j.$$

Using $t = \sum_j D_j$ and rearranging gives the first inequality. Dropping the nonpositive terms with $j \geq 4$ gives the second. □

Let

$$C = v_0 v_1 \cdots v_{t-1} v_0$$

be the outer boundary cycle, oriented counterclockwise. Let θ_i be its interior angle at v_i, and set $\tau_i = \theta_i - \pi$. Thus a positive τ_i is a clockwise turn of the directed boundary. A boundary edge is *triangular* if its endpoints have a common neighbour. Every such common neighbour lies on the interior side of the edge: an equilateral triangle on the outer-face side is empty and would cut off a bounded face there. It is unique, because the two unit circles through the endpoints have only one intersection point on a prescribed side.

A boundary vertex of degree d satisfies $\theta_i \geq (d-1)\pi/3$. We need one additional contribution from every nontriangular boundary edge. For such an edge uv, let ux and vy be the incident edges immediately next to uv on the interior side. The vertices x, u, v, y form a simple three-edge chain of unit edges, and $x \neq y$. Lemma 2.5 says that the two angles adjacent to uv have sum at least π, rather than merely $2\pi/3$.

We now isolate the boundary configurations used in the reduction. A *degree-four run* is a maximal sequence $v_{i+1}, \ldots, v_{i+k}$ of consecutive degree-four boundary vertices such that every edge $v_j v_{j+1}$, $i+1 \leq j < i+k$, is triangular. It is *anchored on the left* if v_i has degree three, $v_i v_{i+1}$ is triangular, and either v_{i-1} has degree three or v_{i-1} has degree four and $v_{i-1}v_i$ is triangular. Right anchoring is defined symmetrically. We call a run anchored at at least one end an *anchored run*. Its *size* is its number of vertices, and it is *concave* if the sum of its turns is at least $\pi/3$.

Call a nontriangular boundary edge, or a boundary vertex of degree five or six, a *boundary defect*. Let b be the number of nontriangular boundary edges, let d_j be the number of boundary vertices of degree j, and put

$$N = b + d_5 + d_6,$$

the number of boundary defects. Let A be the number of anchored runs of size at least eleven, and let B be the number of those runs that are concave.

Lemma 2.7 (Boundary count)**.** *We have*

$$d_3 \geq d_5 + 2d_6 + b + B + 6 \geq N + B + 6.$$

Proof. The baseline contribution at a boundary vertex of degree j is $(j-1)\pi/3$. Every nontriangular edge contributes a further $\pi/3$, by the preceding discussion, and every concave anchored run contributes a further $\pi/3$ through the excess turns of its degree-four vertices. Since distinct maximal degree-four runs are disjoint,

$$(t-2)\pi = \sum_i \theta_i \geq \frac{\pi}{3}\big(2d_3 + 3d_4 + 4d_5 + 5d_6 + b + B\big).$$

Substituting $t = d_3 + d_4 + d_5 + d_6$ and rearranging gives the result. □

The next lemma is the only combinatorial bookkeeping needed to force a long run. Its proof is the specialized $m = 7$ form of the boundary propagation argument in [4].

Lemma 2.8 (Propagation between degree-three vertices). *Let $u = x_0, x_1, \ldots, x_D = v$ be the forward boundary path between two consecutive degree-three vertices, and let x_{D+1} be the next boundary vertex after v. Suppose that the interval from u to v contains no anchored run of size at least eleven and no boundary defect, except possibly the edge x_0x_1 or the vertex x_1. Then at least one of x_{D+1} and vx_{D+1} is a boundary defect.*

Proof. Assume the contrary. Then $\deg(x_{D+1}) \leq 4$ and vx_{D+1} is triangular. All vertices $x_2, \ldots, x_{D-1}$ have degree four, and all boundary edges from x_1x_2 through vx_{D+1} are triangular. Hence there is an anchored run of size at least $D-2$. Our hypothesis gives $D \leq 12$.

Set $k = \lceil D/2 \rceil + 1 \leq 7$, and take the alternating boundary set

$$S = \{x_0, x_2, x_4, \ldots, x_{2\lceil D/2 \rceil}\}.$$

If D is even, the last member is v; if D is odd, it is x_{D+1}. The set is independent. Indeed, suppose that two selected vertices are joined by a chord, and let Q be the cycle formed by that chord and the intervening outer-boundary path. Every internal vertex of the path keeps all of its incident edges in the subgraph induced on or inside Q, since an edge leaving the corresponding Jordan region would cross the chord. A chord endpoint has degree at least two in this subgraph, and it has degree at least three unless it is one of the degree-three vertices u, v, x_{D+1}. Every other boundary vertex of Q has degree at least four, except that an internal occurrence of one of u, v, x_{D+1} has degree three. Consequently

$$\sum_{z \in V(Q)} \big(4 - \deg_H(z)\big) \leq 5,$$

where H is the subgraph induced on or inside Q: each degree-three special vertex contributes at most two when it is a chord endpoint and at most one otherwise, and at most three special vertices occur. This contradicts Lemma 2.6. The same argument, applied to the closed boundary subwalk produced by a coincidence, shows that the selected vertices are distinct.

It remains to count the neighbourhood. Consecutive members of S share the skipped boundary vertex. If D is even, both endpoints of S have degree three and the others have degree four, so

$$|N(S)| \leq (4k-2) - (k-1) = 3k - 1.$$

If D is odd, the sum of the selected degrees is at most $4k-1$. The final two selected vertices share both v and the third neighbour of v, because the two boundary edges at v are triangular and $\deg(v) = 3$. Thus there are at least k repeated neighbour incidences, and again $|N(S)| \leq 3k-1$. This contradicts Corollary 2.2. □

Lemma 2.9. *The outer boundary contains a nonconcave anchored run of size at least eleven.*

Proof. List the degree-three boundary vertices in cyclic order as $u_1, \ldots, u_s$, with $u_{s+1} = u_1$. If every interval from u_j to u_{j+1} contains a boundary defect or a long anchored run, then $N + A \geq s = d_3$. Otherwise relabel so that the interval from u_1 to u_2 contains neither.

For $1 \leq j \leq s$, let N_j and A_j count, respectively, the boundary defects and long anchored runs encountered from u_1 up to u_{j+1}. We claim inductively that

$$N_j + A_j \geq j - 1,$$

and that equality forces the first edge or first vertex after u_{j+1} to be a boundary defect. The claim is true for $j = 1$ by Lemma 2.8. If it holds for $j-1$, then either the inequality is already strict, or equality holds and the first element after u_j is a boundary defect, so passing to the next interval raises $N_j + A_j$ by at least one. If equality persists, the current interval has no long run and no further boundary defect, and Lemma 2.8 forces a boundary defect immediately after u_{j+1}. This proves the claim.

For $j = s$, we obtain $N + A \geq s - 1$. Equality would force a boundary defect immediately after u_1, contrary to the choice of the interval from u_1 to u_2. Hence $N + A \geq d_3$. Combining this with Lemma 2.7 gives $A \geq B + 6$, so at least one long anchored run is nonconcave. □

Choose such a run and, reversing the cyclic labelling if necessary, write its first eleven vertices as

$$p_1, p_2, \ldots, p_{11}.$$

Let p_0 be the anchoring degree-three vertex and p_{-1} its predecessor. Thus

$$\deg(p_0) = 3, \qquad \deg(p_i) = 4 \quad (1 \leq i \leq 11), \qquad \sum_{i=1}^{11} \tau_i < \frac{\pi}{3}, \tag{1}$$

all edges $p_i p_{i+1}$, $0 \leq i \leq 10$, are triangular, and either $\deg(p_{-1}) = 3$, or $\deg(p_{-1}) = 4$ and $p_{-1}p_0$ is triangular. For each $0 \leq i \leq 10$, let q_i be the common neighbour of p_i, p_{i+1} on the interior side of the boundary edge.

3. The forced triangular strip

We first fix the geometry and obtain several separation facts that will later replace repeated planarity arguments.

Lemma 3.1 (Boundary-side orientation)**.** *For every $0 \leq i \leq 10$, the vertex q_i lies on the interior side of the directed boundary edge $p_i p_{i+1}$. Identifying $\mathbb{R}^2$ with $\mathbb{C}$, and writing $\omega = e^{i\pi/3}$, we may therefore choose the orientation so that*

$$q_i = p_i + \omega(p_{i+1} - p_i).$$

Proof. The three vertices form a unit equilateral triangle. If q_i lay on the outer-face side of $p_i p_{i+1}$, the triangle $p_i q_i p_{i+1}$ would bound an empty region: its diameter is one, so no other point can lie in it, and no edge can cross its boundary. That region would be a bounded face on the outer-face side of $p_i p_{i+1}$, contradicting that the edge lies on the outer boundary. Hence q_i is the left equilateral apex of the directed edge. □

Put

$$e_i = p_{i+1} - p_i \qquad (0 \leq i \leq 10).$$

At a degree-four vertex p_j, the turn τ_j is nonnegative, and

$$e_j = e_{j-1} e^{-i\tau_j} \qquad (1 \leq j \leq 10). \tag{2}$$

Every cumulative turn inside the chain is strictly less than $\pi/3$.

Lemma 3.2 (The left endpoint)**.** *Normalize $p_0 = 0$ and $e_0 = 1$. Then $\Re p_{-1} < 1/2$. Consequently, p_{-1} is at distance greater than one from every p_j, q_j, and t_j occurring below with $j \geq 1$.*

Proof. Since $|p_{-1}| = |p_1| = 1$ and $|p_{-1} - p_1| \geq 1$, we have $\Re p_{-1} \leq 1/2$. Equality would make p_{-1} the equilateral apex on the outer-face side of the boundary edge $p_0 p_1$. The empty triangle $p_{-1} p_0 p_1$ would then be a bounded face on that side, contradicting that $p_0 p_1$ is incident with the outer face. Hence $\Re p_{-1} < 1/2$, and in particular $|p_{-1} - p_1| > 1$.

For $j \geq 2$, the forward-cone estimate following from (2) gives

$$\Re p_j > 1 + \frac{j-1}{2} \geq \frac{3}{2}.$$

Likewise, $\Re q_j \geq \Re p_j + 1/2$ for $j \geq 1$, and, whenever t_j is defined, $\Re t_j \geq \Re p_j + 1$. In every case the real part of the difference from p_{-1} is greater than one. □

Lemma 3.3 (Separation along the chain). *For indices in the displayed chain, the following hold.*
(i) If $b-a \geq 2$, then $|p_b - p_a| > 1$.
(ii) If $a \notin \{b, b+1\}$, then $|p_a - q_b| > 1$.
(iii) If $t_b = p_b + \omega(e_b + e_{b+1})$, then $|p_a - t_b| > 1$ for every boundary vertex p_a in the chain.

Proof. Rotate so that the earliest edge occurring in each calculation has direction one. Every later edge has the form $e^{-i\phi}$ with $0 \leq \phi < \pi/3$, and hence has real part greater than $1/2$; every vector $\omega e^{-i\phi}$ has real part at least $1/2$.

For (i), if $b = a+2$, then $|p_b - p_a| = |e_a + e_{a+1}| = 2\cos(\tau_{a+1}/2) > \sqrt{3}$. If $b - a \geq 3$, projection onto e_a already exceeds two.

For (ii), if $a < b$, then $q_b - p_a = e_a + \cdots + e_{b-1} + \omega e_b$, whose projection onto e_a exceeds one. If $a > b+1$, then

$$p_a - q_b = \bar{\omega} e_b + e_{b+1} + \cdots + e_{a-1}.$$

For $a = b+2$, the two unit summands make an angle at most $\pi/3$, so the length is at least $\sqrt{3}$; for larger a, the projection onto e_b exceeds $3/2$.

For (iii), the cases $a < b$, $a = b$, and $a > b+2$ follow immediately by the same projection argument. If $a = b+1$, then

$$p_{b+1} - t_b = \bar{\omega} e_b - \omega e_{b+1},$$

and its squared length is $2 - 2\cos(2\pi/3 - \tau_{b+1}) > 1$. If $a = b+2$, then $p_{b+2} - t_b = \bar{\omega}(e_b + e_{b+1})$, whose length exceeds $\sqrt{3}$. $\square$

We now determine the degree of the vertices q_i toward the next layer.

Lemma 3.4 (Two outer neighbours of q_i). *For every $1 \leq i \leq 9$, the vertex q_i has exactly two neighbours outside*

$$\{p_i, p_{i+1}, q_{i-1}, q_{i+1}\}.$$

Call them r_i, s_i, labelled so that the cyclic order around q_i is

$$s_i, r_i, q_{i-1}, p_i, p_{i+1}, q_{i+1},$$

with non-neighbours among q_{i-1}, q_{i+1} retained only as reference directions.

Proof. First suppose that q_i has three such neighbours a_1, a_2, a_3, encountered in this order in the outer sector from q_{i+1} to q_{i-1}. Put

$$\alpha = \angle a_3 q_i p_i, \qquad \beta = \angle p_{i+1} q_i a_1.$$

The three disjoint angular gaps $a_1 q_i a_2$, $a_2 q_i a_3$, and $p_i q_i p_{i+1}$ are each at least $\pi/3$, so $\alpha + \beta \leq \pi$. Applying Lemma 2.5 to the unit chains q_{i-1}, p_i, q_i, a_3 and $a_1, q_i, p_{i+1}, q_{i+1}$ gives

$$\angle q_i p_i q_{i-1} + \alpha \geq \pi, \qquad \angle q_{i+1} p_{i+1} q_i + \beta \geq \pi.$$

Since the two equilateral sectors at each boundary vertex contribute $2\pi/3$, it follows that

$$\tau_i + \tau_{i+1} = \angle q_i p_i q_{i-1} + \angle q_{i+1} p_{i+1} q_i - \frac{2\pi}{3} \geq \frac{\pi}{3},$$

contrary to (1). Thus there are at most two outer neighbours.

Suppose next that there is at most one. Let

$$B_i = \{p_{i+2}, p_{i-1}, p_{i-3}, \ldots\},$$

where the descending sequence terminates at p_0 when i is odd and at p_{-1} when i is even, and put $S_i = B_i \cup \{q_i\}$. Then $|S_i| = \lfloor i/2 \rfloor + 3 \leq 7$.

The set S_i is independent. The boundary and q_i-separations follow from Lemma 3.3, and the only additional endpoint that can occur is p_{-1}, which is handled by Lemma 3.2.

It remains to count neighbours. When i is odd, the boundary set B_i contains one degree-three vertex and all its other members have degree four. Consecutive members in its descending part share the skipped boundary vertex, so $|N(B_i)| \leq 3|B_i| + 1$. When i is even, either p_{-1} has degree three, giving the same estimate, or p_{-1} has degree four; in the latter case p_{-1} and p_1 share both p_0 and the common third vertex of the two triangular edges, again giving $|N(B_i)| \leq 3|B_i| + 1$.

The possible neighbours $p_i, p_{i+1}, q_{i-1}, q_{i+1}$ of q_i already lie in $N(B_i)$, and by assumption q_i adds at most one further vertex. Hence

$$|N(S_i)| \leq 3|B_i| + 2 = 3|S_i| - 1,$$

contrary to Corollary 2.2. Therefore exactly two outer neighbours exist. $\square$

Whenever

$$s_i = r_{i+1},$$

write $t_i = s_i = r_{i+1}$. The point t_i is the second common unit neighbour of q_i, q_{i+1}, the first being p_{i+1}.

Lemma 3.5 (Strip formula)**.** *If t_i is defined, then*

$$t_i = p_i + \omega(e_i + e_{i+1}).$$

Moreover,

$$\begin{aligned} q_i q_{i+1} \in E(G) &\iff \tau_{i+1} = 0, \\ t_i t_{i+1} \in E(G) &\iff \tau_{i+1} + \tau_{i+2} = 0, \end{aligned}$$

whenever the second expression is defined.

Proof. The two intersections of the unit circles centred at q_i, q_{i+1} are symmetric about the midpoint of $q_i q_{i+1}$. Hence

$$t_i = q_i + q_{i+1} - p_{i+1} = p_i + \omega(e_i + e_{i+1}).$$

Also

$$q_{i+1} - q_i = \bar{\omega} e_i + \omega e_{i+1}.$$

The angle between the two unit summands is $2\pi/3 - \tau_{i+1}$, so their sum has length one exactly when $\tau_{i+1} = 0$. Similarly,

$$t_{i+1} - t_i = \bar{\omega} e_i + \omega e_{i+2},$$

and the angle between the summands is $2\pi/3 - (\tau_{i+1} + \tau_{i+2})$. $\square$

Lemma 3.6 (Two-step separation)**.** *If t_i is defined and the indices lie in the chain, then*

$$|q_{i+2} - t_i| \geq \sqrt{3}.$$

Proof. Write $e_{i+1} = e_i e^{-ia}$ and $e_{i+2} = e_i e^{-i(a+b)}$, where $a, b \geq 0$ and $a + b < \pi/3$. Multiplication by a unit complex number gives

$$|q_{i+2} - t_i| = \left|1 + e^{-ia} + \omega^2 e^{-i(a+b)}\right|.$$

The squared length is $3 + 2H(a,b)$, where

$$H(a,b) = \cos a + \cos(2\pi/3 - b) + \cos(2\pi/3 - a - b).$$

For fixed a,

$$\frac{\partial H}{\partial b} = \sin(2\pi/3 - b) + \sin(2\pi/3 - a - b) \geq 0,$$

so $H(a,b) \geq H(a,0)$. Moreover,

$$2H(a,0) = \cos a + \sqrt{3} \sin a - 1 = 2\cos(a - \pi/3) - 1 \geq 0.$$

Thus the squared distance is at least three. $\square$

Lemma 3.7 (Auxiliary separation)**.** *For all relevant indices,*

$$\begin{aligned} b - a \geq 3 &\implies |q_b - q_a| > 1, \\ t_{i+1} \textit{ defined} &\implies |t_{i+1} - q_i| > 1. \end{aligned}$$

Proof. Rotate so that $e_a = 1$. If $b - a \geq 3$, then

$$q_b - q_a = \bar{\omega} e_a + e_{a+1} + \cdots + e_{b-1} + \omega e_b.$$

Its projection onto e_a is greater than two. Similarly,

$$t_{i+1} - q_i = \bar{\omega} e_i + \omega e_{i+1} + \omega e_{i+2},$$

and its projection onto e_i is at least $3/2$. $\square$

The next cap is the local ingredient used repeatedly later.

Lemma 3.8 (Cap above three consecutive equalities)**.** *Suppose t_a, t_{a+1}, t_{a+2} are defined. Then*

$$\left|N(t_{a+1}) \setminus \{t_a, t_{a+2}, q_{a+1}, q_{a+2}\}\right| \leq 2.$$

Proof. Put $t = t_{a+1}$, $q = q_{a+1}$, and $q' = q_{a+2}$. From the strip formula,

$$q - t = -\omega e_{a+2}, \qquad q' - t = \bar{\omega} e_{a+1},$$

so the smaller angle between these two unit edges is

$$\phi = \angle qtq' = \frac{\pi}{3} + \tau_{a+2} < \frac{2\pi}{3}.$$

No further unit neighbour of t can lie in this smaller sector: its direction would have to be at angular distance at least $\pi/3$ from each endpoint, which would require a sector of width at least $2\pi/3$.

Suppose that three further neighbours occur in the complementary sector. Let x be the first one after q', and z the last one before q, in the cyclic order around t. Since there are at least two gaps between consecutive further neighbours, each of size at least $\pi/3$,

$$\lambda + \rho \leq \frac{4\pi}{3} - \phi, \qquad \lambda = \angle xtq', \quad \rho = \angle qtz.$$

The local cyclic order makes the quadrilaterals t_a, q, t, z and x, t, q', t_{a+2} simple. Lemma 2.5 therefore gives

$$\angle t_a q t + \rho \geq \pi, \qquad \lambda + \angle t q' t_{a+2} \geq \pi.$$

Directly from the edge directions,

$$\angle t_a q t = \frac{\pi}{3} + \tau_{a+1} + \tau_{a+2}, \qquad \angle t q' t_{a+2} = \frac{\pi}{3} + \tau_{a+2} + \tau_{a+3}.$$

Adding the last two inequalities and using the bound on $\lambda + \rho$ yields

$$\frac{2\pi}{3} + \tau_{a+1} + 2\tau_{a+2} + \tau_{a+3} \geq \frac{2\pi}{3} + \phi = \pi + \tau_{a+2}.$$

Hence $\tau_{a+1} + \tau_{a+2} + \tau_{a+3} \geq \pi/3$, contrary to (1). Thus at most two further neighbours exist. $\square$

Lemma 3.9 (No early equality triple)**.** *If t_a, t_{a+1}, t_{a+2} are defined, then $a \geq 4$.*

Proof. It is enough to exclude $a = 1, 2, 3$. In each case Lemma 3.8 gives a cap C of size at most two above t_{a+1}. Consider the following independent sets:

a	S	a fixed set containing $N(S) \setminus C$
1	$\{p_6, q_4, p_3, t_2, q_1, p_0\}$	$\{p_{-1}, p_1, p_2, p_4, p_5, p_7, q_0, q_2, q_3, q_5, q_6, r_1, s_4, t_1, t_3\}$
2	$\{p_7, q_5, p_4, t_3, q_2, p_1, p_{-1}\}$	$\{p_0, p_2, p_3, p_5, p_6, p_8, q_0, q_1, q_3, q_4, q_6, q_7, r_2, s_5, t_2, t_4\} \cup W$
3	$\{p_8, q_6, p_5, t_4, q_3, p_2, p_0\}$	$\{p_{-1}, p_1, p_3, p_4, p_6, p_7, p_9, q_0, q_1, q_2, q_4, q_5, q_7, q_8, r_3, s_6, t_3, t_5\}$.

Here $|W| \leq 2$ accounts for the neighbours of p_{-1} not already listed in the middle row. The boundary–auxiliary separations follow from Lemma 3.3. For each row, Lemma 3.7 separates q_a from q_{a+3} and from t_{a+1}, while Lemma 3.6 separates t_{a+1} from q_{a+3}. Lemma 3.2 handles every pair involving p_{-1}. Thus all three displayed sets are independent. The three fixed sets have sizes 15, at most 18, and 18, respectively. After adding the cap C, the corresponding neighbourhood sizes are at most 17, 20, and 20, equal to $3|S| - 1$ in each case. Corollary 2.2 gives a contradiction. $\square$

It remains to show that the Euclidean strip has at most one break. We first record the planar linkage inequality used in the angle estimate.

Lemma 3.10 (Hinge inequality)**.** *Let A, B, C, D be the vertices, in this order, of a convex quadrilateral. Suppose*

$$|A-B| = |C-D| = 1, \qquad 1 \leq |A-D| \leq 2, \qquad |A-C|, |B-C|, |B-D| \geq 1.$$

Let $\alpha \in [\pi/3, \pi]$ be determined by $|A-D| = 2\sin(\alpha/2)$. Then

$$\angle ABC + \angle BCD \leq \alpha + \frac{2\pi}{3}.$$

Proof. Put $d = |A-D|$, place $A = (0,0)$, $D = (d,0)$, and write

$$x = \angle DAB, \qquad y = \angle CDA.$$

Thus $B = (\cos x, \sin x)$ and $C = (d - \cos y, \sin y)$. The inequalities $|B-D| \geq 1$ and $|A-C| \geq 1$ give $\cos x, \cos y \leq d/2$. Since $d = 2\sin(\alpha/2)$, both x and y are at least

$$L = \arccos(d/2) = \frac{\pi}{2} - \frac{\alpha}{2}.$$

Suppose, contrary to the conclusion, that $x + y < 4\pi/3 - \alpha$. Set $u = (x+y)/2$ and $v = (x-y)/2$. Then $L \leq u < 2\pi/3 - \alpha/2 \leq \pi/2$ and $|v| \leq u - L$. A calculation gives

$$|B-C|^2 = d^2 + 4\cos^2 u - 4d\cos u \cos v.$$

For fixed u, the right-hand side is largest when $|v| = u - L$. At that extremum one of x, y equals L; assume $y = L$. Then C lies on the unit circle about A at polar angle L, while B lies on the same circle at angle $x = 2u - L$. Hence

$$|B-C| = 2\sin(u-L) < 2\sin(\pi/6) = 1,$$

a contradiction. Since the angle sum of the quadrilateral is 2π, the desired inequality follows. □

We now record the two local estimates at a break.

Lemma 3.11 (Break estimates)**.** *If $s_i \neq r_{i+1}$, then*

$$\tau_i + 2\tau_{i+1} + \tau_{i+2} \geq \frac{\pi}{3}, \tag{3}$$

$$\tau_i + \tau_{i+1} \geq \angle q_{i+1}p_{i+1}r_{i+1}. \tag{4}$$

Proof. We use the notation in Figure 1. Let

$$\begin{aligned} \alpha &= \angle q_{i+1}p_{i+1}q_i, & \alpha' &= \angle q_i p_i q_{i-1}, & \alpha'' &= \angle q_{i+2}p_{i+2}q_{i+1},\\ \beta &= \angle p_{i+1}q_i s_i, & \beta' &= \angle r_i q_i p_i, & \gamma &= \angle q_i s_i r_{i+1},\\ \delta &= \angle s_i r_{i+1} q_{i+1}, & \varepsilon &= \angle r_{i+1}q_{i+1}p_{i+1}, & \varepsilon' &= \angle p_{i+2}q_{i+1}s_{i+1}. \end{aligned}$$

Lemma 2.5 gives $\alpha' + \beta' \geq \pi$ and $\alpha'' + \varepsilon' \geq \pi$. Since the angles between the two outer neighbours of each q-vertex are at least $\pi/3$, $\beta' \leq 4\pi/3 - \beta$ and $\varepsilon' \leq 4\pi/3 - \varepsilon$. The angle sum of the pentagon $p_{i+1}q_{i+1}r_{i+1}s_i q_i$ is 3π, and therefore

$$\begin{aligned} \tau_i + 2\tau_{i+1} + \tau_{i+2} &= \alpha' + 2\alpha + \alpha'' - \frac{4\pi}{3}\\ &\geq \alpha - \gamma - \delta + \pi. \end{aligned}$$

The cyclic orders at q_i and q_{i+1} put s_i and r_{i+1} in the same open half-plane bounded by $q_i q_{i+1}$. The unit edges $q_i s_i$ and $q_{i+1} r_{i+1}$ do not cross, by planarity. Hence $q_i, s_i, r_{i+1}, q_{i+1}$ occur in this order on a convex quadrilateral. Lemma 3.10, applied with

$$A = q_i, \quad B = s_i, \quad C = r_{i+1}, \quad D = q_{i+1},$$

gives $\gamma + \delta \leq \alpha + 2\pi/3$. Therefore the preceding estimate implies (3).

For (4), put

$$a = \angle s_i p_{i+1} q_i, \quad h = \angle r_{i+1} p_{i+1} s_i, \quad b = \angle q_{i+1} p_{i+1} r_{i+1}, \quad d = |p_{i+1} - r_{i+1}|.$$

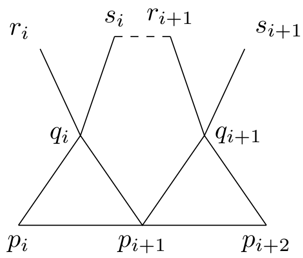

FIGURE 1. The local geometry at a break $s_i \neq r_{i+1}$. The dashed segment is not assumed to be an edge.

The angle at q_i in the isosceles triangle $q_i p_{i+1} s_i$ is at least $\pi/3$, so $0 \le a \le \pi/3$. Then

$$\tau_{i+1} = a + h + b - \frac{\pi}{3}. \tag{5}$$

The isosceles triangle $q_i p_{i+1} s_i$, together with Lemma 2.5 at p_i, q_i, gives

$$\tau_i \ge \frac{\pi}{3} - 2a. \tag{6}$$

If $a \le \pi/6$, the cosine rule in the triangle $s_i p_{i+1} r_{i+1}$, using $|s_i - r_{i+1}| \ge 1$, gives

$$\cos h \le \frac{d^2 - 1 + 4\cos^2 a}{4d\cos a} \le \cos a,$$

because $1 \le d \le 2$ and $d + 1 \le 4\cos^2 a$. Thus $h \ge a$, and adding (5) and (6) yields $\tau_i + \tau_{i+1} \ge b$.

If $a \ge \pi/6$, it is enough by (5) to prove $a + h \ge \pi/3$. The same cosine-rule bound reduces this to

$$d^2 - 1 + (2 - d)(1 + \cos 2a) \le \sqrt{3} d \sin 2a,$$

which follows from $1 \le d \le 2$, $\cos 2a \le 1/2$, and $\sin 2a \ge \sqrt{3}/2$. This proves (4). □

Lemma 3.12 (At most one break)**.** *Among the relations $s_i = r_{i+1}$, $1 \le i \le 8$, at most one fails.*

Proof. Suppose breaks occur at $i < j$. If $i + 2 \le j$, nonnegativity of the turns and (3) give

$$\sum_{h=1}^{11} \tau_h \ge \frac{1}{2}(\tau_i + 2\tau_{i+1} + \tau_{i+2}) + \frac{1}{2}(\tau_j + 2\tau_{j+1} + \tau_{j+2}) \ge \frac{\pi}{3},$$

contrary to (1).

It remains to consider $j = i + 1$. By (4) and its reflected form,

$$\tau_i + \tau_{i+1} + \tau_{i+2} + \tau_{i+3} \ge \angle q_{i+1} p_{i+1} r_{i+1} + \angle s_{i+1} p_{i+2} q_{i+1}.$$

The two triangles with apex q_{i+1} are isosceles, and an angle-sum calculation gives

$$\angle q_{i+1} p_{i+1} r_{i+1} + \angle s_{i+1} p_{i+2} q_{i+1} = \frac{1}{2}\left(\frac{\pi}{3} + \angle s_{i+1} q_{i+1} r_{i+1}\right).$$

Since r_{i+1} and s_{i+1} are distinct unit neighbours of q_{i+1}, their mutual distance is at least one, so the last angle is at least $\pi/3$. Again the total turn is at least $\pi/3$, a contradiction. □

Corollary 3.13 (The unique break)**.** *Exactly one of the relations $s_i = r_{i+1}$, $1 \le i \le 8$, fails, and it is*

$$s_3 \neq r_4.$$

Thus t_i is defined for

$$i \in \{1, 2, 4, 5, 6, 7, 8\}.$$

Moreover, the break inequality is

$$\tau_3 + 2\tau_4 + \tau_5 \ge \frac{\pi}{3}. \tag{7}$$

Proof. By Lemma 3.12, at most one relation fails. Lemma 3.9 forbids the three triples of indices $(1, 2, 3)$, $(2, 3, 4)$, and $(3, 4, 5)$. Hence at least one relation fails, and the unique failure must lie in the intersection of those three triples, namely at index three. Equation (7) is (3) for $i = 3$. □

4. Local elimination

We now contradict minimality in every possible continuation of the strip. We first collect exact neighbourhood information.

Lemma 4.1 (Boundary and endpoint neighbourhoods)**.** *For* $1 \le i \le 10$*,*

$$N(p_i) = \{p_{i-1}, p_{i+1}, q_{i-1}, q_i\}.$$

Also

$$N(p_0) = \{p_{-1}, p_1, q_0\}, \qquad N(p_{11}) \subseteq \{p_{10}, q_{10}\} \cup U_{11}, \quad |U_{11}| \le 2.$$

Finally,

$$\begin{aligned} &N(q_1) \subseteq \{p_1, p_2, q_0, q_2, t_1\} \cup U_1, && |U_1| \le 1, \\ &N(q_9) \subseteq \{p_9, p_{10}, q_8, q_{10}, t_8\} \cup U_9, && |U_9| \le 1. \end{aligned}$$

Proof. The four displayed neighbours of p_i, $1 \le i \le 10$, are distinct by the turn bound and Lemma 3.1; since $\deg(p_i) = 4$, they are all its neighbours. The assertions for p_0, p_{11} follow from their degrees. For q_1 and q_9, Lemma 3.4 gives two outer neighbours. One is already known: $t_1 = s_1$ at q_1, and $t_8 = r_9$ at q_9. At most one remains unknown. □

By Lemma 3.8,

$$\begin{aligned} &|N(t_5) \setminus \{t_4, t_6, q_5, q_6\}| \le 2, \\ &|N(t_6) \setminus \{t_5, t_7, q_6, q_7\}| \le 2, \\ &|N(t_7) \setminus \{t_6, t_8, q_7, q_8\}| \le 2. \end{aligned} \tag{8}$$

The edge t_4t_5.

Lemma 4.2. *If* $t_4t_5 \notin E(G)$*, then*

$$S = \{p_0, p_2, p_4, p_6, p_9, q_7, t_5\}$$

is independent and $|N(S)| \le 20$*.*

Proof. Independence follows from Lemmas 3.3 and 3.6. The selected boundary vertices contribute only

$$\{p_{-1}, p_1, p_3, p_5, p_7, p_8, p_{10}, q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_8, q_9\}.$$

The vertex q_7 adds only t_6, t_7. By (8), the fixed neighbours of t_5 are t_4, t_6, q_5, q_6, and the assumption removes t_4; at most two additional neighbours remain. Hence $|N(S)| \le 18 + 2 = 20$. □

We may therefore assume $t_4t_5 \in E(G)$. Lemma 3.5 gives

$$\tau_5 = \tau_6 = 0. \tag{9}$$

Lemma 4.3. *If* $\tau_7 > 0$*, then*

$$S = \{p_6, p_8, q_4, t_5, t_6, t_7\}$$

is independent and $|N(S)| \le 17$*.*

Proof. Equation (7) and (9) imply $\tau_4 > 0$, so $q_3q_4 \notin E(G)$ by Lemma 3.5. Hence

$$N(q_4) \subseteq \{p_4, p_5, q_5, r_4, t_4\}.$$

The boundary-to-auxiliary separations follow from Lemma 3.3. Normalize $p_4 = 0$ and $e_4 = e_5 = e_6 = 1$, and write $e_7 = e^{-ia}$, $e_8 = e^{-i(a+b)}$, where $a = \tau_7 > 0$, $b \ge 0$. Then

$$\begin{aligned} &q_4 = \omega, && t_5 = 1 + 2\omega, \\ &t_6 = 2 + \omega(1 + e^{-ia}), && t_7 = 3 + \omega(e^{-ia} + e^{-i(a+b)}). \end{aligned}$$

These formulas show directly that q_4 is at distance greater than one from t_5, t_6, t_7. The criterion in Lemma 3.5 shows that t_5t_6 and t_6t_7 are absent, while projection onto the real axis gives $|t_7 - t_5| > 2$. Thus S is independent.

All fixed neighbours of S lie in

$$A = \{p_4, p_5, p_7, p_9, q_5, q_6, q_7, q_8, r_4, t_4, t_8\},$$

which has size at most eleven. The three caps in (8) add at most six vertices, so $|N(S)| \leq 17$. □

Lemma 4.4. *If $\tau_7 = 0$ and $\tau_8 + \tau_9 > 0$, then*

$$S = \{p_0, p_3, p_5, p_8, q_1, q_6, t_7\}$$

is independent and $|N(S)| \leq 20$.

Proof. The boundary separations are covered by Lemma 3.3. Rotate so that $e_1 = 1$. Projection onto the real axis gives

$$\Re(q_6 - q_1) > \frac{5}{2}, \qquad \Re(t_7 - q_1) > \frac{7}{2}.$$

Since $\tau_6 = \tau_7 = 0$, $t_7 - q_6 = e_6 + \omega e_8$, whose length is at least $\sqrt{3}$. Hence S is independent.

The condition $\tau_8 + \tau_9 > 0$ implies $t_7 t_8 \notin E(G)$. Using Lemma 4.1, the exact neighbourhood of q_6, and the cap for t_7, all fixed neighbours lie in

$$A = \{p_{-1}, p_1, p_2, p_4, p_6, p_7, p_9, q_0, q_2, q_3, q_4, q_5, q_7, q_8, t_1, t_5, t_6\}.$$

Thus $|A| \leq 17$, and the unknown sets U_1 and the t_7-cap contribute at most one and two vertices, respectively. Therefore $|N(S)| \leq 20$. □

We are left with the *fully flat* case

$$\tau_5 = \tau_6 = \tau_7 = \tau_8 = \tau_9 = 0. \tag{10}$$

Normalize

$$p_5 = 0, \qquad e_5 = e_6 = e_7 = e_8 = e_9 = 1.$$

Then

$$\begin{array}{lllll} p_6 = 1, & p_7 = 2, & p_8 = 3, & p_9 = 4, & p_{10} = 5, \\ q_5 = \omega, & q_6 = 1 + \omega, & q_7 = 2 + \omega, & q_8 = 3 + \omega, & q_9 = 4 + \omega, \\ t_5 = 2\omega, & t_6 = 1 + 2\omega, & t_7 = 2 + 2\omega, & t_8 = 3 + 2\omega. & \end{array}$$

Also $p_{11} = 5 + e^{-i\theta}$ for some $0 \leq \theta < \pi/3$.

At t_i, $i = 5, 6, 7$, the four forced neighbours occupy directions $0, \pi, 4\pi/3, 5\pi/3$. Every further unit neighbour must therefore lie in the closed angular interval from $\pi/3$ to $2\pi/3$, which we call the *port* of t_i. Let

$$x_i = t_i + \omega, \qquad y_i = t_i + \omega^2$$

be its two endpoints. The port is *saturated* if both endpoint positions are occupied.

Lemma 4.5 (Port geometry)**.** *An unsaturated port contains at most one neighbour. If one endpoint is occupied and the port is not saturated, there is no other port neighbour. Consecutive ports share an endpoint: $x_i = y_{i+1}$.*

Proof. Two unit neighbours in a closed angular interval of width $\pi/3$ have mutual distance at most one. The minimum-distance condition therefore permits two only when they are the endpoints. The identity $x_i = y_{i+1}$ follows from $t_{i+1} = t_i + 1$ and $1 + \omega^2 = \omega$. □

Lemma 4.6 (Degree caps in the flat strip)**.** *If the port of t_7 is saturated, then*

$$N(t_8) \subseteq \{q_8, q_9, t_7, x_7\} \cup U_8, \qquad |U_8| \leq 2.$$

If the ports of t_5, t_6, t_7 are all saturated, then

$$N(x_6) \subseteq \{t_6, t_7, x_5, x_7\} \cup V_6, \qquad |V_6| \leq 2.$$

If the ports of t_6, t_7 are saturated but that of t_5 is not, then

$$N(x_6) \subseteq \{t_6, t_7, y_6, x_7\} \cup V_6, \qquad |V_6| \leq 2.$$

Proof. In the chosen coordinates, the four displayed vertices in each line are distinct unit neighbours of the relevant vertex. The degree bound in Lemma 2.3 leaves at most two further neighbours. □

If the ports of t_5 and t_7 are saturated, then both endpoints of the port of t_6 are occupied, so that port is saturated as well. Thus, apart from the all-saturated case, the possible saturation patterns are

$$\varnothing, \quad \{5\}, \quad \{6\}, \quad \{7\}, \quad \{5,6\}, \quad \{6,7\}.$$

Lemma 4.7 (Unit vectors of the triangular lattice)**.** *For integers a, b,*

$$|a + b\omega|^2 = a^2 + ab + b^2.$$

Consequently, $|a + b\omega| = 1$ if and only if $a + b\omega \in \{\pm 1, \pm\omega, \pm\omega^2\}$.

Proof. The identity follows from $\omega + \bar{\omega} = 1$. If the quadratic form equals one, then

$$1 = \left(a + \frac{b}{2}\right)^2 + \frac{3b^2}{4},$$

so $|b| \le 1$; the three cases $b = -1, 0, 1$ give precisely the six listed vectors. □

Lemma 4.8 (Unsaturated flat configurations)**.** *If the three ports are not all saturated, then G contains an independent set S with $|S| = 6$ and $|N(S)| \le 17$.*

Proof. Only four distinct selected sets are needed:

$$\begin{aligned} I_0 &= \{p_6, p_8, p_{11}, q_9, t_5, t_7\}, \\ I_1 &= \{p_5, p_8, p_{11}, q_6, q_9, t_7\}, \\ I_2 &= \{p_0, p_3, p_6, q_1, q_4, t_5\}, \\ I_3 &= \{p_6, p_9, q_7, t_5, t_8, x_6\}. \end{aligned}$$

The sets I_0, I_1, I_3 are checked in the triangular lattice: every difference between two listed flat-strip points is a nonzero integer combination $a + b\omega$ other than one of $\pm 1, \pm\omega, \pm\omega^2$, the six unit vectors listed in Lemma 4.7. Differences involving $p_{11} = 5 + e^{-i\theta}$ have real part greater than one. The set I_2 is independent by Lemma 3.3; additionally, projection gives $|q_1 - q_4|, |q_1 - t_5| > 1$, and $t_5 - q_4 = 1 + \omega$. Thus all four sets are independent.

For the six possible patterns, define

$$\begin{aligned} A_\varnothing &= \{p_5, p_7, p_9, p_{10}, q_5, q_6, q_7, q_8, q_{10}, t_4, t_6, t_8\}, \\ A_5 &= \{p_4, p_6, p_7, p_9, p_{10}, q_4, q_5, q_7, q_8, q_{10}, t_5, t_6, t_8\}, \\ A_6 &= A_5 \cup \{x_6\}, \\ A_7 &= \{p_{-1}, p_1, p_2, p_4, p_5, p_7, q_0, q_2, q_3, q_5, q_6, r_4, t_1, t_4, t_6\}, \\ A_{56} &= A_5 \cup \{x_6\}, \\ A_{67} &= \{p_5, p_7, p_8, p_{10}, q_5, q_6, q_8, q_9, t_4, t_6, t_7, y_6, x_7\}. \end{aligned}$$

Writing out the neighbourhoods from Lemma 4.1, the three caps in (8), and the port geometry gives the following inclusions. Here C_i denotes the as-yet unlisted port neighbours of t_i.

$$\begin{aligned} &N(I_0) \subseteq A_\varnothing \cup U_{11} \cup U_9 \cup C_5 \cup C_7, && |C_5|, |C_7| \le 1, \\ &N(I_1) \subseteq A_5 \cup U_{11} \cup U_9 \cup C_7 && \text{for the pattern } \{5\}, \\ &N(I_1) \subseteq A_6 \cup U_{11} \cup U_9 && \text{for the pattern } \{6\}, \\ &N(I_2) \subseteq A_7 \cup U_1 \cup C_5, && |C_5| \le 1, \\ &N(I_1) \subseteq A_{56} \cup U_{11} \cup U_9 && \text{for the pattern } \{5,6\}, \\ &N(I_3) \subseteq A_{67} \cup U_8 \cup V_6 && \text{for the pattern } \{6,7\}. \end{aligned}$$

Here $|A_\varnothing|, |A_5|, |A_6|, |A_7|, |A_{56}|, |A_{67}|$ are at most $12, 13, 14, 15, 14, 13$, respectively. Lemmas 4.5 and 4.6 give the stated cap sizes, so every line yields $|N(S)| \le 17$. □

It remains to rule out the all-saturated strip.

Lemma 4.9 (All ports saturated). *If the ports of t_5, t_6, t_7 are all saturated, then*

$$S = \{p_0, p_3, p_6, p_9, q_1, q_4, q_7, t_5, t_8, x_6\}$$

is independent and $|N(S)| \leq 28$.

Proof. The boundary-to-boundary and boundary-to-q, t separations follow from Lemma 3.3. Since $x_6 = p_6 + 3\omega e_6$, it is at distance three from p_6, p_9, while projection along the earliest boundary edge gives $|x_6 - p_3|, |x_6 - p_0| > 1$.

Projection also gives

$$|q_1 - q_4|, |q_1 - q_7|, |q_1 - t_5|, |q_1 - t_8|, |q_1 - x_6| > 1.$$

Equation (7), together with $\tau_5 = 0$, implies $\tau_4 > 0$, so $q_3q_4 \notin E(G)$. In flat coordinates,

$$q_4 = -1 + \omega, \quad q_7 = 2 + \omega, \quad t_5 = 2\omega, \quad t_8 = 3 + 2\omega, \quad x_6 = 1 + 3\omega.$$

Every difference among these five selected vertices is a nonunit triangular-lattice vector by Lemma 4.7. Hence S is independent.

All fixed neighbours lie in

$$A = \{p_{-1}, p_1, p_2, p_4, p_5, p_7, p_8, p_{10}, q_0, q_2, q_3, q_5, q_6, q_8, q_9, \\ r_4, t_1, t_4, t_6, t_7, x_5, x_7, y_5\},$$

which has size at most twenty-three. The endpoint cap U_1 has size at most one, and Lemma 4.6 supplies caps U_8, V_6 of size at most two each. Thus $|N(S)| \leq 23 + 1 + 2 + 2 = 28$. □

5. Proof of the theorem

Proof of Theorem 1.1. Let G be a smallest counterexample. Corollary 3.13 supplies the forced strip described above.

If $t_4t_5 \notin E(G)$, Lemma 4.2 gives an independent set of seven vertices with at most twenty neighbours. If $t_4t_5 \in E(G)$, then $\tau_5 = \tau_6 = 0$. When $\tau_7 > 0$, Lemma 4.3 gives six independent vertices with at most seventeen neighbours. When $\tau_7 = 0$ but $\tau_8 + \tau_9 > 0$, Lemma 4.4 gives seven independent vertices with at most twenty neighbours.

The remaining case is fully flat. If not all three ports are saturated, Lemma 4.8 gives six independent vertices with at most seventeen neighbours. If all are saturated, Lemma 4.9 gives ten independent vertices with at most twenty-eight neighbours.

In every case, $7|N(S)| \leq 20|S|$, contradicting Lemma 2.1. Therefore no counterexample exists, and $\alpha(G) \geq 7|V(G)|/27$. Since the independence number is integral,

$$\alpha(G) \geq \left\lceil \frac{7|V(G)|}{27} \right\rceil.$$

□

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